

ASSIGNMENT: ETHERNET STANDARDS & SPECIFICATIONS

Coursework Documentation & Technical Analysis

Task 1: RJ-45 Connector Pin Layout & Wiring Standards (T568A vs. T568B)

The RJ-45 (Registered Jack 45) connector is the standard physical interface used for twisted-pair Ethernet networking. It features 8 positions and 8 contacts (8P8C). Depending on the application, cables are wired following one of two TIA/EIA wiring standards: T568A or T568B. Below is the pin layout mapping and color coding specification for both standards.

Pin Number	T568A Color Code	T568B Color Code
Pin 1	White / Green	White / Orange
Pin 2	Green	Orange
Pin 3	White / Orange	White / Green
Pin 4	Blue	Blue
Pin 5	White / Blue	White / Blue
Pin 6	Orange	Green
Pin 7	White / Brown	White / Brown
Pin 8	Brown	Brown

Visual Diagram: RJ-45 Pinout Perspective

When looking at the gold pins of the RJ-45 plug facing upward (with the retention clip pointing away/downward), the pins are numbered 1 through 8 from left to right:

Pin 1	Pin 2	Pin 3	Pin 4	Pin 5	Pin 6	Pin 7	Pin 8
W-Org	Org	W-Grn	Blu	W-Blu	Grn	W-Brn	Brn

Note: T568A and T568B are electrically interchangeable in modern networking, provided both ends of a cable use the same standard (straight-through). Mixing them across standard patch cables results in a crossover configuration.

Task 2: Comparison of Ethernet Standards

Ethernet technology has evolved significantly over decades to support higher bandwidth demands. The table below compares three foundational twisted-pair Ethernet standards: 10BASE-T, 100BASE-TX, and 1000BASE-T.

Standard	Speed / Bandwidth	Cable Type	Max Cable Length	Connector Type
10BASE-T	10 Mbps	Cat 3 (or better) Twisted-Pair	100 meters (328 ft)	RJ-45
100BASE-TX	100 Mbps (Fast Ethernet)	Cat 5 (or better) Twisted-Pair	100 meters (328 ft)	RJ-45
1000BASE-T	1,000 Mbps (1 Gbps / Gigabit)	Cat 5e / Cat 6 Twisted-Pair	100 meters (328 ft)	RJ-45

Technical elaboration:

- 10BASE-T: Utilizes Manchester encoding over two pairs of unshielded twisted-pair (UTP) copper cabling.
- 100BASE-TX: Employs 4B5B block coding with MLT-3 signaling over two pairs of Cat 5 UTP cables.
- 1000BASE-T: Uses all four twisted pairs simultaneously in both directions via five-level pulse amplitude modulation (PAM-5) and sophisticated hybrid echo cancellation circuits.

Task 3: Direct Laptop-to-Laptop Ethernet Connection

Scenario: You need to connect two laptops directly using Ethernet without an intervening network switch or router. Which cable (straight-through or crossover) would you use and why?

Analysis & Recommendation:

1. Modern Auto-MDIX Standard (Recommended Approach):

If both laptops were manufactured within the last two decades (supporting 1000BASE-T Gigabit Ethernet or modern 100BASE-TX controllers), you can use a standard

If both laptops are modern devices, **straight-through cable** can be used because modern network interface cards (NICs) incorporate **Auto-MDIX (Automatic Medium-Dependent Interface Crossover)** technology. Auto-MDIX automatically detects the polarity of the signals on the cable and electronically swaps the transmit (TX) and receive (RX) pairs inside the hardware chipset, rendering physical crossover cables unnecessary.

Traditional/Legacy Hardware Context (Crossover Cable):

If connecting older legacy equipment (such as 10BASE-T or early 100BASE-TX hardware without Auto-MDIX), a **crossover cable** is required. In traditional Ethernet wiring, devices are categorized as DTE (Data Terminal Equipment, like computers and routers) and DCE (Data Communication Equipment, like switches and hubs). DTE devices transmit on pins 1 & 2 and receive on pins 3 & 6. When connecting two DTE devices (laptop to laptop) directly with a straight-through cable, both devices would transmit on pin 1 & 2 and listen on pins 3 & 6, resulting in a collision of transmission signals with no receiver attached. A crossover cable physically crosses the wire pairs—connecting Pin 1 (TX+) on one end to Pin 3 (RX+) on the other end, and Pin 2 (TX-) to Pin 6 (RX-)—thereby aligning the transmitter of Laptop A with the receiver of Laptop B.

Task 4: Real-World Ethernet Devices & Standards

Three real-world devices commonly found in home or college environments that utilize wired Ethernet connections, along with their supported Ethernet standards, are detailed below:

1. Network Attached Storage (NAS) Enclosure / Home Media Server

Supported Standard: 1000BASE-T (Gigabit Ethernet) / 2.5GBASE-T

Operational Role: NAS units (such as Synology or QNAP enclosures) rely heavily on wired Ethernet for high-speed file transfers, streaming 4K media, and network backups. Gigabit Ethernet (1000BASE-T) provides up to 1000 Mbps of bidirectional throughput, preventing bottlenecks during simultaneous multi-user file access.

2. Smart TV / 4K Streaming Media Player (e.g., Apple TV 4K)

Supported Standard: 100BASE-TX (Fast Ethernet) or 1000BASE-T (Gigabit Ethernet)

Operational Role: While many smart TVs use Wi-Fi, high-end models and dedicated streaming boxes include an RJ-45 port. Standard HD streaming requires only 5–15 Mbps, easily satisfied by 100BASE-TX, but modern 4K HDR streaming and uncompressed audio passthrough benefit significantly from 1000BASE-T ports to eliminate buffering.

3. Wi-Fi 6 / Wi-Fi 7 Wireless Access Point (WAP) or Enterprise Router

Supported Standard: 1000BASE-T or 2.5GBASE-T (Multi-Gigabit Ethernet)

Operational Role: Modern wireless access points deployed in college campuses or smart homes aggregate massive wireless client traffic. Because modern Wi-Fi speeds exceed standard 1 Gbps wireless backhaul capacity, these access points connect via Gigabit Ethernet (1000BASE-T) or Multi-Gigabit (2.5GBASE-T) to prevent the wired uplink from becoming the network bottleneck.